

Thermodynamic Properties of the Stirling Engine

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August 2024

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1 Introduction

Varying the environment that a Stirling heat engine is subject to appears to cause variations in the relative effectiveness of its heat transfer mechanisms. We may attempt to explain these variations using thermodynamic theory. In a Gamma arrangement Stirling Engine Cycle there are two pistons connected to each other. One is a displacer piston, which exists to push cold and hot air to one side of the piston and assist the heating/cooling of the air. The power piston models a sealed piston and is moved by the heating and cooling of the air in the piston. A depiction of this engine is included in Figure 5 in the appendix (Salev 2021), followed by images of our setup. In this report, we aim to vary the torque on the flywheel and make observations on how this affects the operation of the engine. We must also change the supplied voltage to ensure our heat reservoir remains constant temperature. Some interesting findings that we encounter are the decrease in output power at high RPM, and strong correlations between decreased efficiency at high RPM. We will find that these arise due to our ideal assumptions being applied to a non-ideal engine.

2 Results and Discussion

In order to determine a relationship between torque, power, and efficiency with the speed of the engine (in RPM) it is imperative that we maintain the temperature of the reservoirs. If the reservoirs change temperature, it will affect these values in such a way that is difficult to isolate and account for.

As we increased the torque on the engine by hand-adjusting the tension strap on the engine's exterior (pictured in the appendix), we needed to decrease the supplied voltage so as to ensure the temperature of the hot reservoir was maintained at $560 \pm 5^\circ C$.

Once the temperature of the hot reservoir reached a stable value within this range, we were able to take recordings of the instantaneous hot and cold reservoir temperatures, the engine RPM, the force applied on the engine in Newtons, and the voltage and current supplied to the engine.

Stipulating that the formula for torque is

$$\tau_{\text{engine}} = F_{\text{band}} \times r_{\text{wheel}}, \text{ where } r = 0.02\text{m}, \quad (1)$$

we can construct a plot of torque versus engine speed.

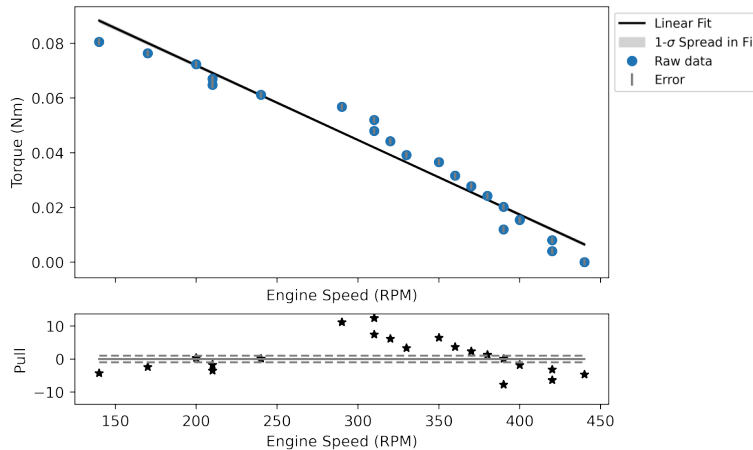


Figure 1: Graph of torque against engine speed with linear fit line. Second plot shows the number of standard errors each data point is from the linear fit line, giving us an indication of the data's precision and accuracy.

As we increase the torque on the engine, the speed of the engine decreases. This is an expected and relatively intuitive result, insofar as that if you apply greater opposing force on a rotating object, it will cause it to rotate slower.

We may also create a plot of the power produced by the Stirling engine, or the shaft power. In Figure 2, we see that as the speed of the engine increases, we have an initial increase in the produced power, followed by a plateau, and then a decrease in the power produced.

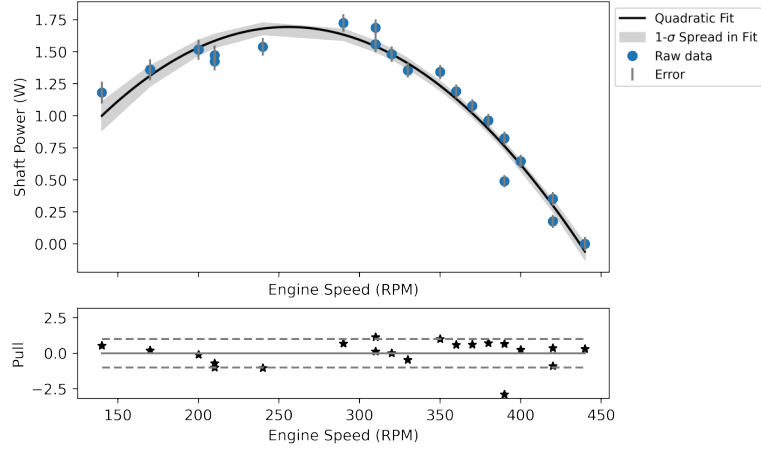


Figure 2: Graph of shaft power against engine speed with quadratic fit line. Second plot shows the number of standard errors each data point is from the fit line.

This is quite an interesting result. The intuitive result would be that there is a linear correlation between increasing RPM and increasing shaft power. The explanation for this change in correlation comes from our incorrect assumption that the Stirling engine is ideal. We assumed that

$$P_{\text{shaft}} = \tau_{\text{engine}} \times 2\pi N \text{ rps} \quad (2)$$

for all N and τ . In reality, at higher RPM, the effects of friction increase much faster and the quasi-equilibrium assumption starts to break down. The engine cannot take in enough air to continue operating close to our ideal conditions. The assumption works for lower RPMs, as the quasi-equilibrium assumption assumes infinitely slow processes. The resulting peak is called the power band¹ (Muir 2024). A possible work around for this would be making use of gears, like we see in cars.

The next information we can analyse is the supplied heat power to the engine. This is defined as

$$P = I \times V. \quad (3)$$

The input power is this power, minus the power required to maintain the system at $560 \pm 5^\circ\text{C}$.

$$P_{\text{input}} = I \times V - I_0 \times V_0. \quad (4)$$

Figure 2 shows input power against engine RPM.

¹The power band is not necessarily quadratic, as pictured. It refers to the subsection of the entire plot encompassing the RPM-range with the highest power output.

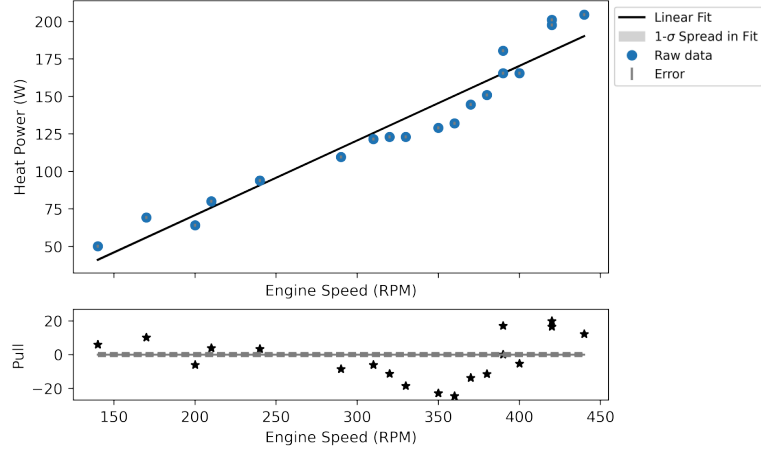


Figure 3: Graph of heat power against engine speed with linear fit line. Second plot shows the number of standard errors each data point is from the linear fit line.

In this instance, we see a direct linear correlation between an increase in supplied power and engine speed. This is an expected result. Since we are not measuring any output of the engine here, we are not suffering the side effects of the engine being non-ideal. The input power will make the engine run faster, which is an intuitive and sensical result.

The efficiency is defined as

$$\eta_{th} = \frac{W_{out}}{\dot{Q}_{in}}, \quad (5)$$

which is plotted against engine speed in Figure 4.

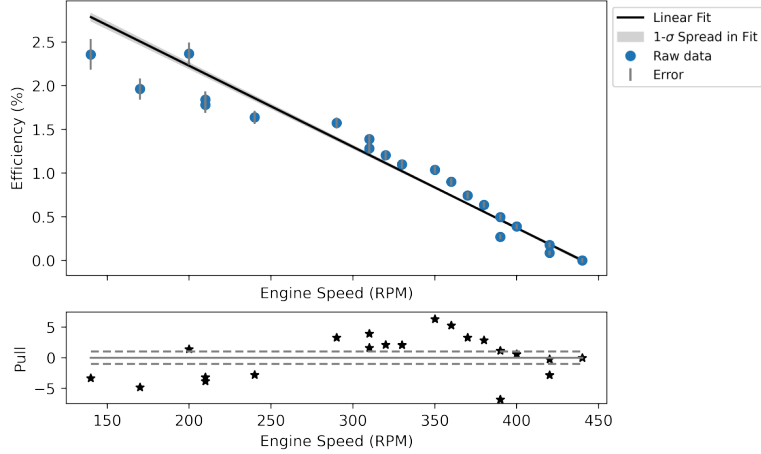


Figure 4: Graph of efficiency against engine speed with linear fit line. Second plot shows the number of standard errors each data point is from the linear fit line.

We find that efficiency decreases as engine speed increases. The engine efficiency is far lower than the theoretical ideal efficiency for a Stirling engine. Some features of the real Stirling engine that we don't take into account are that the engine has lots of heat loss due to friction, imperfect insulation, radiation, conduction between surfaces, and non-instantaneous heat transfer. Air is also not an ideal gas. All these factors that we have not taken into account in the theoretical efficiency of the engine would mean that the actual efficiency is far, far lower than theorised. The decrease in efficiency at higher RPM can also be attributed to the increase in friction (heat loss) at high RPM, and also the limitations of the non-instantaneous compression of air (non-ideal gas).

These incorrect assumptions also create large amounts of error within our data. We can see that our data points tend to vary in how many standard uncertainties they are from the fit line, due to the rapid changes in thermodynamic variables that we have not accounted for. We see data points many sigma² from the fit line as a result.

Another major source of error is the imperfect experimental setup. As mentioned above, we are dealing with imperfect sealings in reality. We also have effects of radiation and conduction of heat energy to other surfaces, resulting in heat loss. In a more ideal experimental setup, these variables would be more controlled by using materials with higher heat capacities, for example.

3 Application

A Stirling engine that is designed to mitigate much of this imperfection and heat loss could theoretically be used to produce power on satellites and Mars missions. One example is the Advanced Stirling Radioisotope Generator (ASRG) by NASA Glenn. For this problem, we are told that the engine must produce 3.6kJ of work ($W_b = 3600\text{J}$). It has a max capacity of 2.5L (0.0025m^3), a compression ratio of 2.5, and max working pressure of 10MPa, and will use air as the working fluid. The hot side must be kept at 400°C .

We will take state 1 to be the compressed state, where $P_1 = 10\text{MPa}$, $V_1 = \frac{0.0025}{2.5} = 0.001\text{m}^3$, and $T_1 = T_2 = 673.15\text{K}$. The isobars tell us that $V_1 = V_4$ and $V_2 = V_3$. The initial expanded volume is also $V_3 = 0.0025\text{m}^3$.

The assumptions we need to make are that air is a calorically perfect and ideal gas, the Stirling cycle is ideal (quasi-equilibrium), the pistons are massless, there is no friction, and the system is perfectly sealed and insulated.

A sketch of the P-v diagram for this cycle is provided in Figure 8 in the appendix.

In an isothermal process, which occurs for states 1 to 2, and 3 to 4, $PV = \text{constant}$ by definition.

$$\begin{aligned} C_1 &= P_1 V_1 \\ C_1 &= 2.5 \times 10^4 \\ C_2 &= P_3 V_3 \\ C_2 &= mRT_3, \text{ where } R_{\text{air}} = 287\text{JK}^{-1} \end{aligned} \tag{6}$$

So, $P_{12} = \frac{C}{V_1}$ and $P_{34} = \frac{mRT_3}{V_3}$. To determine the required temperature of the cold reservoir, let's take the produced 3.6kJ of work to be the ideal work produced. Boundary work is produced in the isothermal processes, as isobaric processes produce no work

$$\begin{aligned} W &= \int_{V_1}^{V_2} PdV + \int_{V_3}^{V_4} PdV \\ &= \int_{V_1}^{V_2} PdV + \int_{V_2}^{V_1} PdV \\ &= \int_{V_1}^{V_2} (P_{12} - P_{34})dV \end{aligned} \tag{7}$$

Substituting our values leaves us with

$$\begin{aligned} W &= (2.5 \times 10^4 - 287mT_3) \int_{V_1=0.001}^{V_2=0.0025} \frac{1}{V} dV \\ W &= 0.916(2.5 \times 10^4 - 287mT_3) \end{aligned} \tag{8}$$

The total energy, as stipulated, is 3.6kJ. Solving for T_3 ³, we get $T_3=567.9\text{K}$. This is the cold side of the engine. The ideal efficiency is thus

$$\begin{aligned} \eta_{\text{th}} &= 1 - \frac{T_L}{T_H} \\ \eta_{\text{th}} &= 15.64\% \end{aligned} \tag{9}$$

²One sigma refers to one standard error. The pull graphs show how many sigma each data point is from the fit line.

³We know the value of mR since $P_1 V_1 = mRT_1$.

15.7% of the heat input is being used to create work. The rest of it is radiated as heat loss to maintain the cold temperature. The radiated energy is

$$\begin{aligned} E_{\text{rad}} &= (1 - 0.1564) * \frac{3600}{0.1563} \\ E_{\text{rad}} &= 19.4\text{kJ} \end{aligned} \tag{10}$$

To find the size of the radiating surface, we may use the Stefan-Boltzmann Law:

$$\frac{P}{A} = \sigma e(T^4 - T_C^4), \tag{11}$$

where $P = 19.4\text{kW}$, $e = 0.96$ (given), $T = 2.7\text{K}$, and $T_C = 567.9\text{K}$. Hence, the area is 3.43m^2 .

Hydrogen would be a much better choice of working fluid. For this application, we have considered air as an ideal gas. In reality, this is not the case. An ideal gas requires no collisions between particles within the gas. Since hydrogen has a much smaller mean molecular weight than air, and less atomic particles, it would be a much better approximation of an ideal gas.

4 Conclusion

We found that varying the torque on a Gamma type Stirling engine had a great effect on the thermodynamic variables of the system. Increasing the torque has a direct correlation with decreased RPM. This relationship is linear, as described in Figure 1. Interestingly, when exploring the output power as a function of engine RPM, we found that, while there was an initial predicted rise in power output with RPM, we reached a limit where the power output started to decrease with RPM. Such a phenomenon arises since our assumption of a quasi-equilibrium process breaks down for fast compression and expansion. We also found that as the engine operated faster, we needed to provide more heat input to maintain its operation. This is an intuitive result. And lastly, we found that efficiency decreased for higher RPM. This occurs for the same reason as for the power output; our idealised assumptions start to break down at higher RPM as we have not taken into account heat loss due to friction and fast compression/expansion at these speeds.

Future experiments may like to explore how changing the working fluid affects these results. For example, using hydrogen instead of air (to achieve more idealised results). One may also extend the aims of an experiment like this to explore how non-ideal gases operate in Stirling engines.

5 References

- K. Salev (2021). *How Gamma Type Stirling Engine Works*. [online]. Available at [weblink](#).
A. Muir (2024). *Torque and BHP Explained*, How A Car Works. [online]. Available at [weblink](#).

6 Appendices

6.1 Appendix 1

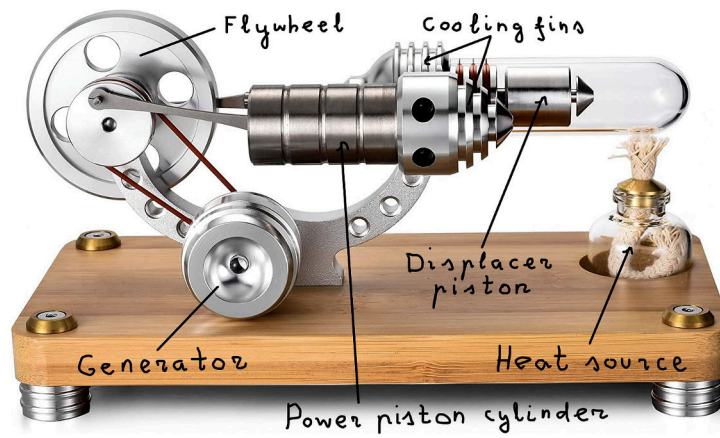


Figure 5: Labelled apparatus of Gamma type Stirling engine (Salev 2021).



Figure 6: Image taken of Stirling engine setup. On the bottom is the hot reservoir and the displacer piston, and then further up we see the power piston on the right, and the flywheel on the left. The torque is varied by adjusting the tension on the strap pictured on the left of the flywheel.

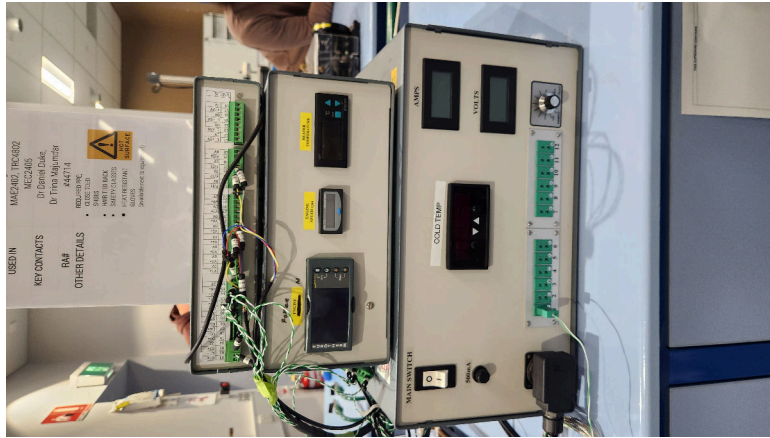


Figure 7: Image taken of recording devices. This set up allows us to record the hot and cold reservoir temperatures, the current and voltage, the engine RPM, and the applied force.

6.2 Appendix 2

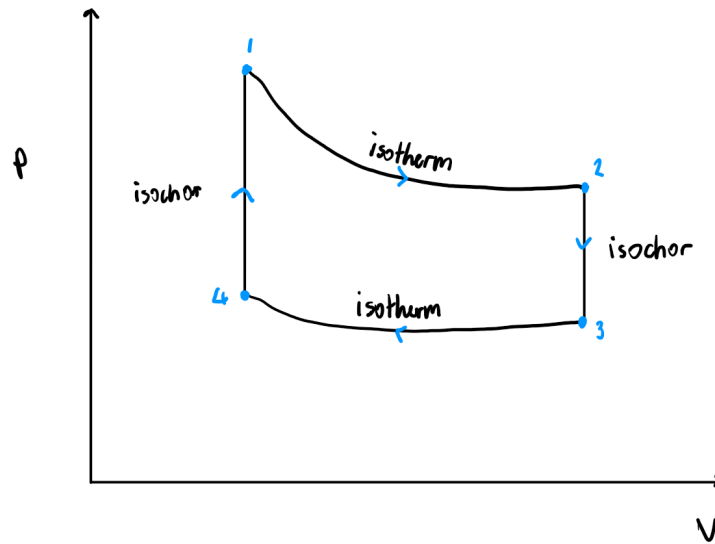


Figure 8: Sketch of idealised P-V diagram for Stirling cycle.